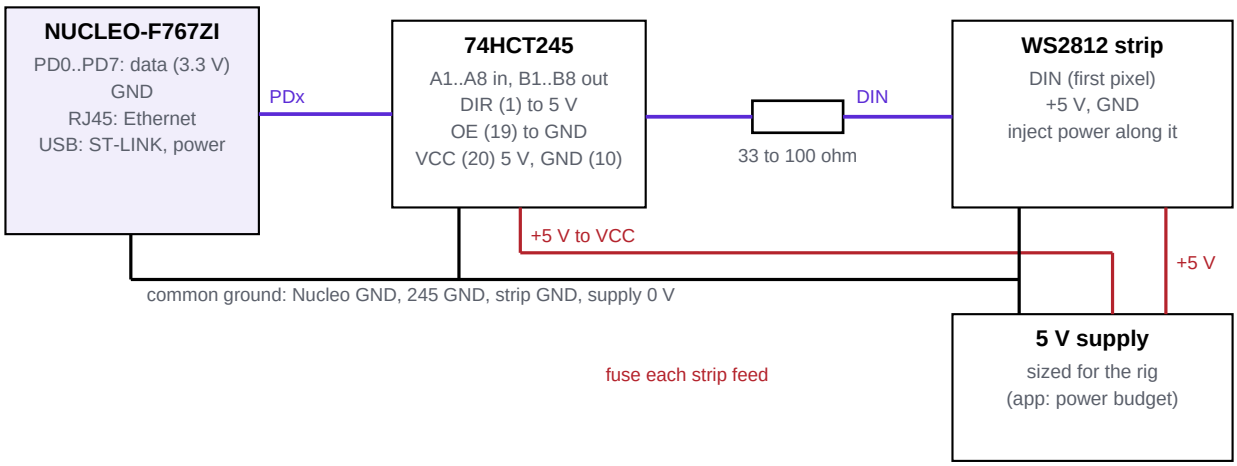
Psyight pixel node

Wiring, pinout, flashing and set-up for the NUCLEO-F767ZI running `firmware/node-f767` (stage F3, config protocol 3, data on port D).

What	Where
Data lines (8)	PD0 to PD7, all on the Zio headers (seven on CN9, one on CN8), one WS2812-class strip chain each, through a 74HCT245
Network	RJ45 (CN14), 100 Mbit; Art-Net (UDP 6454) and sACN; config on UDP 6455
Programming and console	USB on the ST-LINK side (CN1): flashing, and a virtual COM port at 115200 8N1
Power for the board	USB (ST-LINK) for the bench; the strips never draw from the Nucleo

1. One data line, end to end



The MCU's outputs are 3.3 V; WS2812 inputs want at least 0.7 x VDD (3.5 V at 5 V), so each line goes through a 74HCT245 powered at 5 V (HCT inputs accept 3.3 V as high). A 33 to 100 ohm series resistor at the 245's output damps ringing on the cable. Keep the data cable short (a few metres) or mount the node near the strip; for long runs use RS-485 per line (not built yet).

2. Data line pins

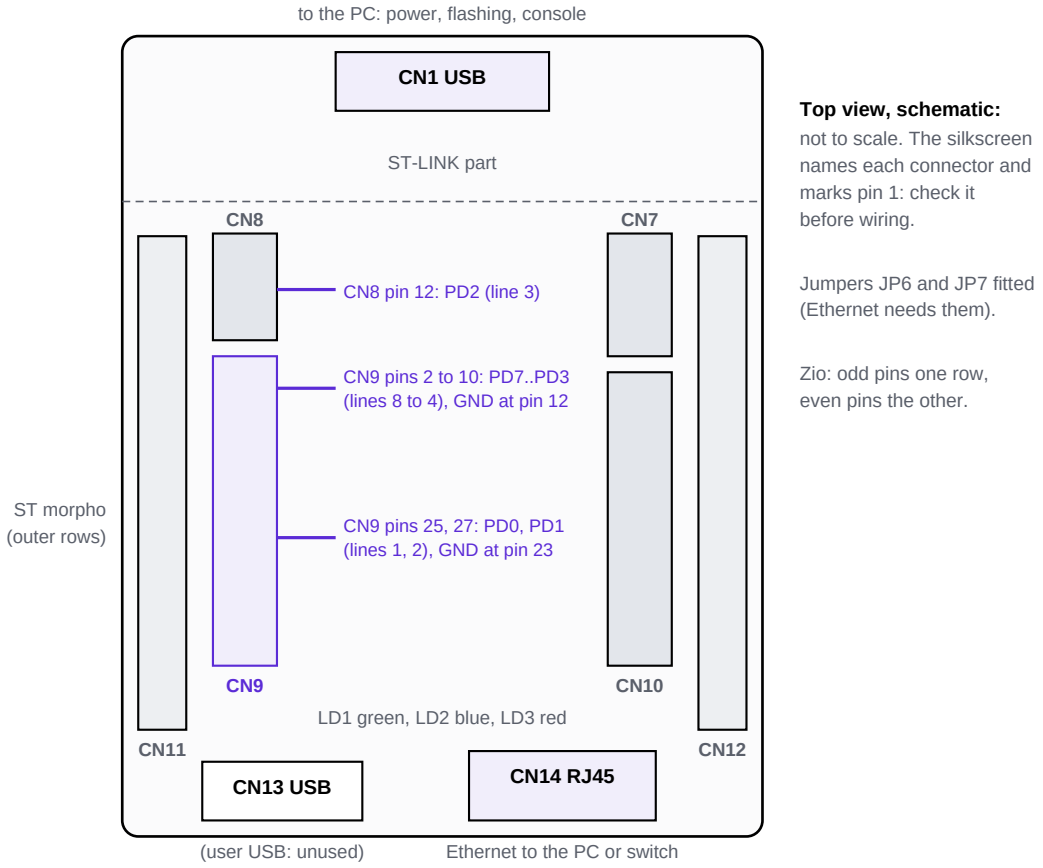
All eight lines sit on the Zio headers. The ST morpho column is the same pin on the long outer header, if that suits the harness better: use one or the other for a pin, not both. The app's Preflight > Node setup numbers lines 1 to 8; line n is lane n-1 and pin PD(n-1).

Line (app)	Lane	STM32 pin	Zio connector	ST morpho	Zio name
1	0	PD0	CN9 pin 25	CN11 pin 57	D67
2	1	PD1	CN9 pin 27	CN11 pin 55	D66
3	2	PD2	CN8 pin 12	CN11 pin 4	D48
4	3	PD3	CN9 pin 10	CN11 pin 40	D55
5	4	PD4	CN9 pin 8	CN11 pin 39	D54
6	5	PD5	CN9 pin 6	CN11 pin 41	D53
7	6	PD6	CN9 pin 4	CN11 pin 43	D52
8	7	PD7	CN9 pin 2	CN11 pin 45	D51

- Lines 4 to 8 (PD3 to PD7) are side by side on CN9 pins 10, 8, 6, 4 and 2, with a ground at CN9 pin 12 next to them: one 6-way connector carries five lines and their ground.
- Lines 1 and 2 (PD0, PD1) are on CN9 pins 25 and 27, a ground at CN9 pin 23; line 3 (PD2) is on CN8 pin 12.

- No board function uses PD0 to PD7 (ST's board file). PD8 and PD9 on the same port carry the console; the output only ever drives bits 0 to 7.
- The default configuration drives lines 1 to 4 (the app's 2.5 by 1.2 m test frame, `active 15`); Node setup sets the lines the current rig needs.
- Firmware before 2026-10-23 used PE0 to PE7: flash the current F3 before wiring port D.

3. Board pinout



The Zio headers, pin by pin (UM1974 Table 13; the Zio names follow each pin). Odd pins in the left column, even pins in the right. PA7 and PB13 carry Ethernet (with JP6 and JP7 fitted): leave them free. PB0 is the green LED.

CN8 (Zio, left, upper)

1	NC	2	PC8 D43
3	IOREF	4	PC9 D44
5	RESET	6	PC10 D45
7	+3.3V	8	PC11 D46
9	+5V	10	PC12 D47
11	GND	12	PD2 D48 line 3
13	GND	14	PG2 D49
15	VIN	16	PG3 D50

CN7 (Zio, right, upper)

1	PC6 D16	2	PB8 D15
3	PB15 D17	4	PB9 D14
5	PB13 D18 Ethernet	6	AREF
7	PB12 D19	8	GND
9	PA15 D20	10	PA5 D13
11	PC7 D21	12	PA6 D12
13	PB5 D22	14	PA7 D11 Ethernet
15	PB3 D23	16	PD14 D10
17	PA4 D24	18	PD15 D9
19	PB4 D25	20	PF12 D8

CN9 (Zio, left, lower)

1	PA3 A0	2	PD7 D51 line 8
3	PC0 A1	4	PD6 D52 line 7
5	PC3 A2	6	PD5 D53 line 6
7	PF3 A3	8	PD4 D54 line 5
9	PF5 A4	10	PD3 D55 line 4
11	PF10 A5	12	GND
13	NC D72	14	PE2 D56
15	PA7 D71 Ethernet	16	PE4 D57
17	PF2 D70	18	PE5 D58
19	PF1 D69	20	PE6 D59
21	PF0 D68	22	PE3 D60
23	GND	24	PF8 D61
25	PD0 D67 line 1	26	PF7 D62
27	PD1 D66 line 2	28	PF9 D63
29	PG0 D65	30	PG1 D64

CN10 (Zio, right, lower)

1	AVDD	2	PF13 D7
3	AGND	4	PE9 D6
5	GND	6	PE11 D5
7	PB1 A6	8	PF14 D4
9	PC2 A7	10	PE13 D3
11	PF4 A8	12	PF15 D2
13	PB6 D26	14	PG14 D1
15	PB2 D27	16	PG9 D0
17	GND	18	PE8 D42
19	PD13 D28	20	PE7 D41
21	PD12 D29	22	GND
23	PD11 D30	24	PE10 D40
25	PE2 D31	26	PE12 D39
27	GND	28	PE14 D38
29	PA0 D32	30	PE15 D37
31	PB0 D33 LD1	32	PB10 D36
33	PE0 D34	34	PB11 D35

4. Power, ground and the other pins the node uses

Signal	Zio	ST morpho	Notes
+5 V	CN8 pin 9	CN11 pin 18	Board 5 V. Do not feed strips from it.
+3.3 V	CN8 pin 7	CN11 pin 16	Not needed by the 245 (it runs at 5 V).
GND	CN8 pins 11, 13; CN9 pins 12, 23; CN10 pins 5, 17, 22, 27; CN7 pin 8	CN11 pins 8, 19, 20, 22, 49, 60; CN12 pins 9, 20, 39, 54, 63	Tie to the 245's GND and the strip supply's 0 V.
LD1 green	PB0 (CN10 pin 31)		Blinks at 1 Hz while the firmware runs.
LD2 blue	PB7		On while show data arrives.
LD3 red	PB14		On without an Ethernet link, or after a fault.
Console	USART3: PD8 TX, PD9 RX		Wired to the ST-LINK: the USB virtual COM port, 115200 8N1.
Ethernet (RMII)	PA1, PA2, PA7, PB13, PC1, PC4, PC5, PG11, PG13		Onboard LAN8742A PHY. Jumpers JP6 and JP7 must be fitted (closed) for Ethernet.
Timer and DMA	TIM1, DMA2 (three streams into GPIOD BSRR)		Internal: no pins. TIM1's own pins (PE9, PE11, PE13) stay free.

5. 74HCT245 connections (DIP-20 or SO-20)

245 pin	Name	Connect to
1	DIR	+5 V (A to B)
2 to 9	A1 to A8	PD0 to PD7 (line 1 on A1 ... line 8 on A8)
10	GND	Common ground
18 down to 11	B1 to B8	Series resistor, then strip DIN (B1 for line 1 ... B8 for line 8)
19	OE (active low)	GND (always enabled)
20	VCC	+5 V, with a 100 nF capacitor to GND at the pin

6. Power and safety

- One ground for everything: the Nucleo, the 245 and every strip supply's 0 V.
- Fuse each output (each strip feed) for its cable and the strip's rating (SPEC 12.2, 13).
- Inject 5 V along long strips: a WS2812 draws up to about 60 mA at full white, so 150 pixels can draw 9 A. The app's power budget and brightness cap keep the show within the supply.

- Power the strips before (or with) the node, never data into an unpowered strip.
- The flash limiter in the app is on by default; it is a safeguard, not a certification.

7. Building and flashing the firmware

Needs STM32CubeIDE 1.19 (its arm-none-eabi-gcc 13.3 is found automatically), CMake 3.25 or later and Ninja. From `firmware/node-f767`:

```
powershell -ExecutionPolicy Bypass -File fetch_deps.ps1
cmake -S . -B build/f3 -G Ninja -DCMAKE_TOOLCHAIN_FILE=%CD%/cmake/arm-none-eabi.cmake
-DNODE_STAGE=3
cmake --build build/f3
```

Plug the Nucleo's ST-LINK USB (CN1) into the PC, then flash with STM32CubeProgrammer's command line (installed with STM32CubeIDE or on its own):

```
STM32_Programmer_CLI -l st-link
STM32_Programmer_CLI -c port=SWD -w build/f3/psyight_node_f3.elf -v -rst
```

- The first command lists the probe (it should say NUCLEO-F767ZI).
- The second erases only the sectors the program needs, writes, verifies and resets. The node's saved settings (the last flash sector) are kept; settings saved by protocol 2 firmware are read and upgraded at start.
- Without the command line: copy `psyight_node_f3.bin` onto the NODE_F767ZI USB drive the ST-LINK shows.
- Console: open the ST-LINK virtual COM port at 115200 8N1. Every 5 s it prints the address, output frames per second, packets, latches, sequence gaps, ignored universes, and "signal lost" while no show data arrives.

8. Setting the node up from the app

- Connect the node to the same network as the PC, or straight to the PC's Ethernet port. It asks for an address by DHCP; without a lease (a cable straight to the PC) it takes a link-local 169.254.x.y/16 address from its MAC after 10 s, the range Windows gives the PC on the same cable, so they find each other with no settings. The console prints it. A fixed address: set dhcp 0 with ip and mask (by default the Art-Net style 2.x.y.z/8).
- In Psyight: Preflight > Discover Art-Net nodes, then Node setup. It shows the data lines the current rig needs (pixels and colour order per line), keeping each line at or under 545 pixels (60 fps) when the eight lines allow it.
- Choose what the node does when the signal stops (hold, fade, fallback show or blackout), after how long, and the fallback show's colours, brightness and speed. These are saved with the project.
- Set up the node: the app sends everything and applies it at once (no restart), then reads it back and lists any difference. Save on the node keeps it across restarts (output pauses 1 to 2 s while the flash sector erases).

Text protocol (UDP 6455), for scripts or a terminal: get, set <key> <value>, apply, save, defaults, reboot. Keys: universe, universes, pixels_per_universe, lane0 to lane7 (first pixel and count), order0 to order7 (rgb, grb, brg, rbg, gbr, bgr), active (line bit mask), loss_ms, policy, fade_ms, fallback_brightness and fallback_speed (thousandths), palette (one to four RRGGBB), name, long_name, protocols, ip, mask, gateway, dhcp.

9. Frame rate per line

Pixels per line	Universes	Frame time	Max frame rate
170	1	5.4 ms	about 185 fps
340	2	10.5 ms	about 95 fps
510	3	15.6 ms	about 64 fps
545	3.2	16.7 ms	60 fps (the show's rate)
680 (the most a line takes)	4	20.7 ms	about 48 fps

WS2812: 30 microseconds a pixel plus a 300 microsecond reset; all eight lines run in parallel, so the longest line sets the rate. Example: the Lotus preset (1,427 pixels) needs three lines: seed and inner petals 1 to 5 (537 pixels), inner 6 to 8 and outer 1 to 3 (510), outer 4 to 8 (380).

10. Bench checklist

- Scope PD0 (CN9 pin 25) with F1: bits 1.25 us (plus or minus 150 ns), 0 high 0.40 us, 1 high 0.80 us, frame gap at least 280 us; repeat with heavy network traffic.
- Link at 100 Mbit full duplex; unplug and replug: the node must come back.
- sACN multicast through the switch (IGMP snooping); every line changes on the same frame (sync universe 1000 or ArtSync).
- Stop sending: after loss_ms the chosen policy runs; check the fallback show's colours on each line (colour order per line, protocol 3).
- Set up from the app, apply without a restart (no glitch mid-frame), save, power cycle, read back.
- If the red LED stays on or the address stays 0.0.0.0: check the cable, then that JP6 and JP7 are fitted.

Sources: ST UM1974 Rev 7, Tables 11, 13 and 21 (NUCLEO-F767ZI); the STM32CubeMX board file for the NUCLEO-F767ZI; Psyight firmware sources (src/ws2812, src/board, src/net, src/core/node_config.h); SPEC sections 12 and 13. Generated by tools/python/make_node_guide.py.